



Can a small detector tell us who is not wearing a hardhat?

PPE compliance detection on construction sites • YOLOv8

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The check that happens on foot

Anchor project: DDIP grain processing facility, Djibouti (AHG portfolio)

A safety officer walks the site to check PPE. They see one area at a time, and the record is a paper form.

We asked a narrow question: can a detector trained on public data flag a worker missing a hardhat or hi-vis vest in a site photo?

It is decision support, not enforcement. The model flags; a person decides.

Recall on NO-Hardhat

target ≥ 0.70

a missed violation is the costly error

Overall mAP50

target ≥ 0.60

realistic for a 30-epoch nano model

Credentials to reproduce

zero

runs in Colab from the public repo

Success criteria, set before training

Public data, and the rights that come with it

Construction Site Safety (Roboflow Universe) • CC BY 4.0 • 5 of 10 classes kept

The dataset

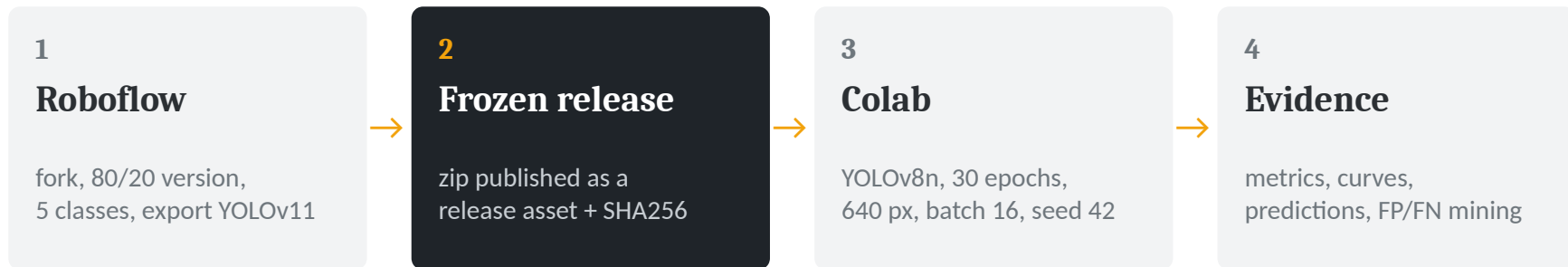
- Classes: Hardhat • NO-Hardhat • Safety Vest • NO-Safety Vest • Person
- 717 source images • 574 train / 143 valid (80 / 20)
- Preprocessing: auto-orient, resize 640×640 , no augmentation
- 20 of the upstream 25 classes dropped — cones, plant, vehicles, masks

What publishing it obliged us to do

- Republishing a dataset is redistribution — so the release carries the upstream CC BY 4.0 licence and its attribution
- No client or employer photographs anywhere: the repository and its release are public
- Training images show real workers. A licence to reuse an image is not consent from the people in it, so no face crops are published
- Ultralytics YOLOv8 is AGPL-3.0 — commercial use would need compliance or an enterprise licence

Frozen data, so the result can be re-run

Roboflow annotates and versions · the release is what the notebook reads



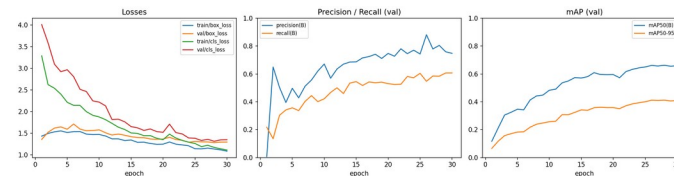
Why not download from Roboflow in the notebook?

That needs an API key, which makes reproduction depend on the reader having an account — and on Roboflow still serving that exact version later. Our notebook downloads a frozen zip over plain HTTPS and verifies its SHA256 before training, so anyone can re-run it with no credentials at all. A wrong checksum stops the run instead of training on the wrong data.

Results

Validation split · 143 images · confidence 0.25 · IoU 0.50 · 30 epochs in 5.4 min on a T4

Class	Precision	Recall	mAP50	mAP50-95
all	0.806	0.584	0.662	0.411
Hardhat	0.952	0.633	0.760	0.486
NO-Hardhat	0.700	0.510	0.545	0.291
NO-Safety Vest	0.767	0.499	0.573	0.338
Person	0.832	0.626	0.716	0.485
Safety Vest	0.779	0.653	0.717	0.456



mAP, recall and both losses were still improving at epoch 30 — the run stopped while the model was still learning, so it is undertrained rather than saturated.

mAP50 0.662 clears our 0.60 target. NO-Hardhat recall 0.510 misses our 0.70 target. Precision beats recall on every class: the model is cautious, and the two classes carrying the safety signal are its weakest.

Where it fails, and why that matters

128 false positives and 251 false negatives — but reading the images changed what they mean

False negatives — a violation goes unflagged

- A worker behind scaffolding poles: the vest was found at 0.83, the body was not
- A navy jacket with reflective bands not counted as a vest, though the hardhat above it was found at 0.98
- Crowd scenes: the model fires on some people and not others, regardless of size

False positives — a false alarm to check

- 4 of the 6 worst come from one shopping-mall photo — bare heads called at 0.85–0.95
- Every one of them is correct. The annotator labelled some heads and stopped
- The sixth is real: a plain lime t-shirt called Safety Vest at 0.84 — bright colour, no banding

It learned colour, not banding — and the dataset is not really a construction dataset

A bright t-shirt reads as a vest; a banded navy jacket does not. Almost every vest in training is fluorescent, so the model learned hue. And with retail queues, selfies and stock photography in the data, it has partly learned 'bare head anywhere' rather than 'bare head on a site' — so the headline figure measures two domains at once.

What we would fix first

Data before models — a bigger model mostly learns the existing gaps more confidently

1

Break the colour bias in the vest classes

Non-fluorescent workwear and bright non-workwear, plus a decision on banding vs hue · check: the lime-shirt and navy-jacket pair stops reproducing

2

Add ~200 occluded, crowded and rear-view site images

Targets the scaffolding and crowd misses · check: Person recall rises from 0.626 to 0.750 and misses fall below 40

3

Filter out the non-construction images and complete the labels

Retail queues, selfies, stock photography · check: NO-Hardhat precision rises from 0.700 to 0.820

Limitations we are not claiming past: two PPE items only · untested on any AHG site · weak on occluded and crowded scenes · one validation split of 143 images · Safety Vest rests on just 49 instances.

Check it yourself in three steps

No account, no API key, no secrets

Open

Click the Colab badge in the
README

Reset

Runtime → Disconnect and delete
runtime

Run

Run all · watch for SHA256 verified

Repository

github.com/solomon8909/maicen0526-m4u3-ppe-detection

Release v1.0 holds the frozen dataset, the unseen images and the trained weights, each with a checksum.

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